

1.2 Gaussian Elimination

This section develops a systematic procedure for solving linear systems: row operations that reduce the augmented matrix to a form whose solution can be read off by inspection.

Considerations in Solving Linear Systems

Solving linear systems by hand (small systems) differs from solving them by computer (large systems, sometimes with millions of unknowns), which must also manage memory, roundoff error, and solution time – topics of **numerical analysis**, touched on only briefly here. Even so, computer methods for large systems build on the ideas developed in this section.

Echelon Forms

In Example 6 of Section 1.1 we reduced an augmented matrix to

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

from which $x = 1, y = 2, z = 3$ was immediate. This is a matrix in **reduced row echelon form**, meaning:

1. Each nonzero row's first nonzero entry is a 1, called a **leading 1**.
2. Zero rows are grouped at the bottom.
3. Each leading 1 lies strictly to the right of the leading 1 in the row above.
4. Every column containing a leading 1 is zero elsewhere.

A matrix with only the first three properties is in **row echelon form** (reduced row echelon form is a special case).

EXAMPLE 1 | Row Echelon and Reduced Row Echelon Form

Reduced row echelon form:

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Row echelon form but not reduced:

$$\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

EXAMPLE 2 | More on Row Echelon and Reduced Row Echelon Form

Row echelon form has zeros only *below* each leading 1; reduced row echelon form has zeros below *and above*. With any values for the *'s, these are row echelon form:

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

These are reduced row echelon form:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Once row operations reduce an augmented matrix to *reduced* row echelon form, the solution set is read off directly or converted to parametric form:

EXAMPLE 3 | Unique Solution

Suppose the augmented matrix for a system in x_1, x_2, x_3, x_4 reduces to

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix}$$

This gives $x_1 = 3$, $x_2 = -1$, $x_3 = 0$, $x_4 = 5$ – a unique solution, i.e. the 4-tuple $(3, -1, 0, 5)$.

EXAMPLE 4 | Linear Systems in Three Unknowns

Solve the system in x, y, z whose augmented matrix reduces to each of the following.

$$(a) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution (a) The last row gives $0x + 0y + 0z = 1$ – inconsistent.

Solution (b) The last row ($0 = 0$) can be dropped, leaving

$$x + 3z = -1, \quad y - 4z = 2$$

Variables tied to leading 1's (x, y) are **leading variables**; the rest (z) are **free variables**. Solving for the leading variables:

$$x = -1 - 3z, \quad y = 2 + 4z$$

Setting $z = t$ gives the parametric equations

$$x = -1 - 3t, \quad y = 2 + 4t, \quad z = t$$

E.g., $t = 0$ gives $(-1, 2, 0)$; $t = 1$ gives $(-4, 6, 1)$.

Solution (c) Dropping the zero rows leaves the single equation

$$x - 5y + z = 4 \tag{1}$$

– a plane in three-dimensional space. Solving for x : $x = 4 + 5y - z$; setting $y = s$, $z = t$ gives

$$x = 4 + 5s - t, \quad y = s, \quad z = t \tag{2}$$

Such parametric formulas have a name:

Definition 1

For a system with infinitely many solutions, parametric equations that generate every solution by assigning numerical values to the parameters form a **general solution** of the system.

Formula (2) is thus a general solution of system (c) above.

Elimination Methods

Once a matrix is in reduced row echelon form, solving is easy. The following step-by-step algorithm reduces *any* matrix to this form, illustrated on

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Step 1. Locate the leftmost column that is not entirely zero.

$$\begin{bmatrix} \mathbf{0} & 0 & -2 & 0 & 7 & 12 \\ \mathbf{2} & 4 & -10 & 6 & 12 & 28 \\ \mathbf{2} & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad (\text{leftmost nonzero column: column 1})$$

Step 2. If needed, interchange rows so this column's top entry is nonzero.

$$\begin{bmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad (\text{rows 1 and 2 interchanged})$$

Step 3. If that top entry is a , multiply row 1 by $1/a$ to make it a leading 1.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad (\text{row 1 multiplied by } \tfrac{1}{2})$$

Step 4. Add multiples of the top row to the rows below to zero out everything beneath the leading 1.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix} \quad (-2 \times \text{row 1 added to row 3})$$

Step 5. Cover the top row and repeat Steps 1–4 on the remaining submatrix, until the whole matrix is in row echelon form.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix} \quad (\text{submatrix row 1 multiplied by } -\tfrac{1}{2})$$

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \end{bmatrix} \quad (-5 \times \text{submatrix row 1 added to submatrix row 2})$$

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (\text{new submatrix's only row multiplied by 2})$$

The matrix is now in row echelon form. One more step gives reduced row echelon form.

Step 6. Working upward from the last nonzero row, add multiples of each row to the rows above to zero out everything above the leading 1's.

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (\tfrac{7}{2} \times \text{row 3 added to row 2})$$

$$\begin{bmatrix} 1 & 2 & -5 & 3 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (-6 \times \text{row 3 added to row 1})$$

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad (5 \times \text{row 2 added to row 1})$$

This last matrix is in reduced row echelon form.

This algorithm is **Gauss–Jordan elimination**: a **forward phase** (zeros below the leading 1's) followed by a **backward phase** (zeros above). Using only the forward phase yields a row echelon form and is called **Gaussian elimination** – as obtained above at the end of Step 5.

EXAMPLE 5 | Gauss–Jordan Elimination

Solve by Gauss–Jordan elimination.

$$\begin{aligned}x_1 + 3x_2 - 2x_3 & \quad + 2x_5 & = 0 \\2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 & = -1 \\5x_3 + 10x_4 & \quad + 15x_6 & = 5 \\2x_1 + 6x_2 & \quad + 8x_4 + 4x_5 + 18x_6 & = 6\end{aligned}$$

SOLUTION: The augmented matrix is

$$\left[\begin{array}{ccccccc}1 & 3 & -2 & 0 & 2 & 0 & 0 \\2 & 6 & -5 & -2 & 4 & -3 & -1 \\0 & 0 & 5 & 10 & 0 & 15 & 5 \\2 & 6 & 0 & 8 & 4 & 18 & 6\end{array}\right]$$

Add $-2 \times$ row 1 to rows 2 and 4:

$$\left[\begin{array}{ccccccc}1 & 3 & -2 & 0 & 2 & 0 & 0 \\0 & 0 & -1 & -2 & 0 & -3 & -1 \\0 & 0 & 5 & 10 & 0 & 15 & 5 \\0 & 0 & 4 & 8 & 0 & 18 & 6\end{array}\right]$$

Multiply row 2 by -1 , then add $-5 \times$ (new row 2) to row 3 and $-4 \times$ (new row 2) to row 4:

$$\left[\begin{array}{ccccccc}1 & 3 & -2 & 0 & 2 & 0 & 0 \\0 & 0 & 1 & 2 & 0 & 3 & 1 \\0 & 0 & 0 & 0 & 0 & 0 & 0 \\0 & 0 & 0 & 0 & 0 & 6 & 2\end{array}\right]$$

Interchange rows 3 and 4, then multiply the new row 3 by $\frac{1}{6}$ – the row echelon form:

$$\left[\begin{array}{ccccccc}1 & 3 & -2 & 0 & 2 & 0 & 0 \\0 & 0 & 1 & 2 & 0 & 3 & 1 \\0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\0 & 0 & 0 & 0 & 0 & 0 & 0\end{array}\right]$$

Forward phase complete. Add $-3 \times$ row 3 to row 2, then $2 \times$ (new row 2) to row 1 – the reduced row echelon form:

$$\left[\begin{array}{ccccccc}1 & 3 & 0 & 4 & 2 & 0 & 0 \\0 & 0 & 1 & 2 & 0 & 0 & 0 \\0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\0 & 0 & 0 & 0 & 0 & 0 & 0\end{array}\right]$$

Backward phase complete; the corresponding system is

$$x_1 + 3x_2 \quad + 4x_4 + 2x_5 \quad = 0, \quad x_3 + 2x_4 \quad = 0, \quad x_6 = \frac{1}{3} \quad (3)$$

Solving for the leading variables:

$$x_1 = -3x_2 - 4x_4 - 2x_5, \quad x_3 = -2x_4, \quad x_6 = \frac{1}{3}$$

Assigning the free variables x_2, x_4, x_5 the parameters r, s, t gives the general solution:

$$x_1 = -3r - 4s - 2t, \quad x_2 = r, \quad x_3 = -2s, \quad x_4 = s, \quad x_5 = t, \quad x_6 = \frac{1}{3}$$

Homogeneous Linear Systems

A linear system is **homogeneous** if all constant terms are zero:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0 \end{aligned}$$

Every homogeneous system is consistent, since $x_1 = \cdots = x_n = 0$ (the **trivial solution**) always works; any other solution is **nontrivial**.

So only two outcomes are possible:

- The system has only the trivial solution.
- The system has infinitely many solutions in addition to the trivial solution.

For two homogeneous equations in two unknowns,

$$a_1x + b_1y = 0 \text{ } [a_1, b_1 \text{ not both zero}], \quad a_2x + b_2y = 0 \text{ } [a_2, b_2 \text{ not both zero}],$$

both graphs are lines through the origin, and the trivial solution is their common point there (**Figure 1.2.1**).

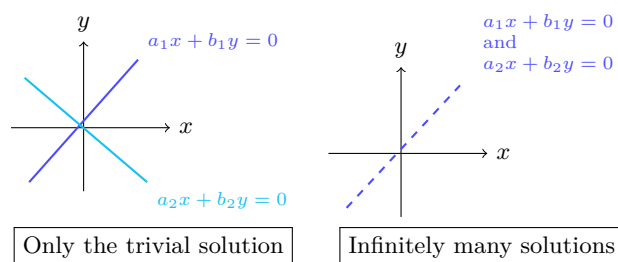


FIGURE 1.2.1

One case guarantees nontrivial solutions: more unknowns than equations. Consider four equations in six unknowns.

EXAMPLE 6 | A Homogeneous System

Use Gauss–Jordan elimination to solve the homogeneous linear system

$$\begin{aligned}x_1 + 3x_2 - 2x_3 & \quad \quad + 2x_5 & = 0 \\2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 & = 0 \\5x_3 + 10x_4 & \quad \quad + 15x_6 & = 0 \\2x_1 + 6x_2 & \quad \quad + 8x_4 + 4x_5 + 18x_6 & = 0\end{aligned}\tag{4}$$

SOLUTION: This is Example 5’s system with all right-hand sides set to zero, so its reduced row echelon form matches Example 5’s except the last column is all zeros (row operations never change a zero column):

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding system is

$$x_1 + 3x_2 \quad + 4x_4 + 2x_5 = 0, \quad x_3 + 2x_4 = 0, \quad x_6 = 0$$

giving $x_1 = -3x_2 - 4x_4 - 2x_5$, $x_3 = -2x_4$, $x_6 = 0$. With $x_2, x_4, x_5 = r, s, t$, the parametric solution is

$$x_1 = -3r - 4s - 2t, \quad x_2 = r, \quad x_3 = -2s, \quad x_4 = s, \quad x_5 = t, \quad x_6 = 0$$

(The trivial solution is $r = s = t = 0$.)

Free Variables in Homogeneous Linear Systems

Example 6 illustrates two points about homogeneous systems:

1. Row operations never change a zero column, so the reduced row echelon form of a homogeneous system’s augmented matrix keeps its zero last column – the reduced system stays homogeneous.
2. A zero row corresponds to $0 = 0$, which imposes no condition and can be dropped – so the reduced system may have fewer equations than the original.

In general, for n unknowns and r nonzero rows in the reduced row echelon form, there are r leading variables and $n - r$ free variables.

Theorem 1.2.1 – Free Variable Theorem for Homogeneous Systems

If a homogeneous linear system has n unknowns, and if the reduced row echelon form of its augmented matrix has r nonzero rows, then the system has $n - r$ free variables.

If $m < n$ (fewer equations than unknowns), then $r < n$, so there is at least one free variable – hence infinitely many solutions:

Theorem 1.2.2

A homogeneous linear system with more unknowns than equations has infinitely many solutions.

So Example 6's system (4 equations, 6 unknowns) was guaranteed to have infinitely many solutions.

Gaussian Elimination and Back-Substitution

Gauss–Jordan elimination suits hand computation; for large computer-solved systems, Gaussian elimination (row echelon form) plus **back-substitution** is typically more efficient:

EXAMPLE 7 | Example 5 Solved by Back-Substitution

From Example 5's row echelon form

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

solve the corresponding system

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0, \quad x_3 + 2x_4 + 3x_6 = 1, \quad x_6 = \frac{1}{3}$$

as follows:

Step 1. Solve the equations for the leading variables.

$$x_1 = -3x_2 + 2x_3 - 2x_5, \quad x_3 = 1 - 2x_4 - 3x_6, \quad x_6 = \frac{1}{3}$$

Step 2. Starting from the bottom, substitute each equation into those above it.

Substituting $x_6 = \frac{1}{3}$ into the second equation:

$$x_1 = -3x_2 + 2x_3 - 2x_5, \quad x_3 = -2x_4, \quad x_6 = \frac{1}{3}$$

Substituting $x_3 = -2x_4$ into the first equation:

$$x_1 = -3x_2 - 4x_4 - 2x_5, \quad x_3 = -2x_4, \quad x_6 = \frac{1}{3}$$

Step 3. Assign arbitrary values to the free variables, if any.

Assigning $x_2, x_4, x_5 = r, s, t$ gives the general solution

$$x_1 = -3r - 4s - 2t, \quad x_2 = r, \quad x_3 = -2s, \quad x_4 = s, \quad x_5 = t, \quad x_6 = \frac{1}{3}$$

matching Example 5.

EXAMPLE 8 | Existence and Uniqueness of Solutions

Each matrix below is a row-echelon (not reduced) augmented matrix for a system in x_1, x_2, x_3, x_4 . Discuss existence and uniqueness of solutions.

$$(a) \begin{bmatrix} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & -3 & 7 & 2 & 5 \\ 0 & 1 & 2 & -4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Solution (a) Last row: $0x_1 + 0x_2 + 0x_3 + 0x_4 = 1$ – inconsistent.

Solution (b) Last row: $0 = 0$, no effect. x_1, x_2, x_3 are leading variables, x_4 is free – infinitely many solutions.

Solution (c) Last row gives $x_4 = 0$; back-substituting into $x_3 + 6x_4 = 9$ gives $x_3 = 9$, and continuing upward pins down x_2 then x_1 uniquely – a unique solution.

Some Facts About Echelon Forms

Three facts worth knowing (not proved here):

1. Every matrix has a unique reduced row echelon form, regardless of which sequence of row operations is used.
2. Row echelon forms, however, are not unique – different operation sequences can give different results.
3. Still, every row echelon form of a matrix A has the same number of zero rows and the same leading-1 positions, called the **pivot positions** of A ; the columns/rows containing them are the **pivot columns/pivot rows**, and a nonzero entry in a pivot position is a **pivot** of A .

EXAMPLE 9 | Pivot Positions and Columns

Earlier we reduced

$$A = \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad \text{to} \quad \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

The leading 1's sit at (row 1, col 1), (row 2, col 3), (row 3, col 5) – the pivot positions. So the pivot columns are 1, 3, 5; the pivot rows are 1, 2, 3; and the pivots are the entries there.

Roundoff Error and Instability

Theory and practice can diverge: computers approximate numbers, introducing **roundoff** error that, uncontrolled, can make an algorithm **unstable**. Since Gauss–Jordan elimination takes roughly 50% more operations than Gaussian elimination on large systems, most computer algorithms use the latter.

Exercise Set 1.2

1. Classify each matrix as row echelon form, reduced row echelon form, both, or neither.

$$(a) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & 4 \end{bmatrix} \quad (e) \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(f) \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (g) \begin{bmatrix} 1 & -7 & 5 & 5 \\ 0 & 1 & 3 & 2 \end{bmatrix}$$

3. Each augmented matrix below is a row echelon form reached by row operations. Identify its pivot rows/columns and solve the system.

$$(a) \begin{bmatrix} 1 & -3 & 4 & 7 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & 7 & -2 & 0 & -8 & -3 \\ 0 & 0 & 1 & 1 & 6 & 5 \\ 0 & 0 & 0 & 1 & 3 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & -3 & 7 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. Solve by Gaussian elimination.

$$x_1 + x_2 + 2x_3 = 8, \quad -x_1 - 2x_2 + 3x_3 = 1, \quad 3x_1 - 7x_2 + 4x_3 = 10$$

7. Solve by Gaussian elimination.

$$x - y + 2z - w = -1, \quad 2x + y - 2z - 2w = -2, \quad -x + 2y - 4z + w = 1, \quad 3x - 3w = -3$$

9. Solve Exercise 5 by Gauss-Jordan elimination.

11. Solve Exercise 7 by Gauss-Jordan elimination.

13. Determine by inspection (no pencil and paper) whether the homogeneous system has nontrivial solutions.

$$2x_1 - 3x_2 + 4x_3 - x_4 = 0, \quad 7x_1 + x_2 - 8x_3 + 9x_4 = 0, \quad 2x_1 + 8x_2 + x_3 - x_4 = 0$$

15. Solve by any method.

$$2x_1 + x_2 + 3x_3 = 0, \quad x_1 + 2x_2 = 0, \quad x_2 + x_3 = 0$$

17. Solve by any method.

$$3x_1 + x_2 + x_3 + x_4 = 0, \quad 5x_1 - x_2 + x_3 - x_4 = 0$$

21. Solve by any method.

$$2I_1 - I_2 + 3I_3 + 4I_4 = 9, \quad I_1 - 2I_3 + 7I_4 = 11, \quad 3I_1 - 3I_2 + I_3 + 5I_4 = 8, \quad 2I_1 + I_2 + 4I_3 + 4I_4 = 10$$

23. Each augmented matrix below has unspecified entries *. Determine whether the system is consistent and, if so, whether the solution is unique – answer “inconclusive” if there isn’t enough information.

$$(a) \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{bmatrix} \quad (b) \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & * & * & * \\ 0 & 0 & * & 0 \\ 0 & 0 & 1 & * \end{bmatrix}$$

25. Determine the values of a for which the system has no solution, exactly one solution, or infinitely many solutions.

$$x + 2y - 3z = 4, \quad 3x - y + 5z = 2, \quad 4x + y + (a^2 - 14)z = a + 2$$

27. What condition, if any, must a, b, c satisfy for the system to be consistent?

$$x + 3y - z = a, \quad x + y + 2z = b, \quad 2y - 3z = c$$

31. Find two different row echelon forms of

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

– showing that row echelon forms need not be unique.

32. Reduce

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 3 & 4 & 5 \end{bmatrix}$$

to reduced row echelon form without introducing fractions at any intermediate stage.

33. Show that the following nonlinear system has 18 solutions for $0 \leq \alpha, \beta, \gamma \leq 2\pi$.

$$\sin \alpha + 2 \cos \beta + 3 \tan \gamma = 0, \quad 2 \sin \alpha + 5 \cos \beta + 3 \tan \gamma = 0, \quad -\sin \alpha - 5 \cos \beta + 5 \tan \gamma = 0$$

[Hint: substitute $x = \sin \alpha, y = \cos \beta, z = \tan \gamma$.]

35. Solve the following nonlinear system for x, y, z .

$$x^2 + y^2 + z^2 = 6, \quad x^2 - y^2 + 2z^2 = 2, \quad 2x^2 + y^2 - z^2 = 3$$

[Hint: substitute $X = x^2, Y = y^2, Z = z^2$.]

37. Find a, b, c, d so that the curve shown is the graph of $y = ax^3 + bx^2 + cx + d$.

39. If the linear system

$$a_1x + b_1y + c_1z = 0, \quad a_2x - b_2y + c_2z = 0, \quad a_3x + b_3y - c_3z = 0$$

has only the trivial solution, what can be said about the solutions of

$$a_1x + b_1y + c_1z = 3, \quad a_2x - b_2y + c_2z = 7, \quad a_3x + b_3y - c_3z = 11 ?$$

40. (a) If A has three rows and five columns, what is the maximum possible number of leading 1’s in its reduced row echelon form?

(b) If B has three rows and six columns, what is the maximum possible number of parameters in the general solution of the system with augmented matrix B ?

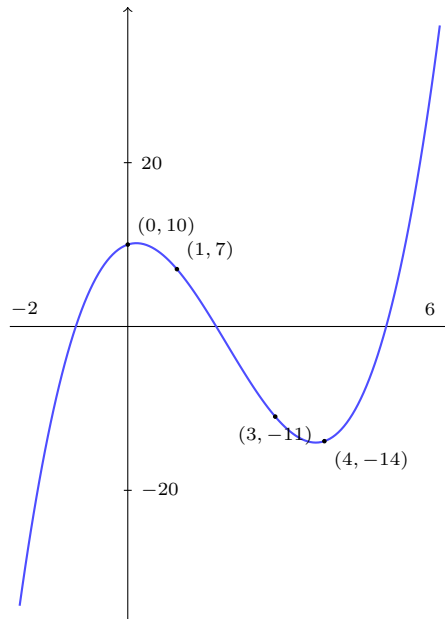


FIGURE Ex-37

(c) If C has five rows and three columns, what is the minimum possible number of zero rows in any row echelon form of C ?

41. Describe all possible reduced row echelon forms of

$$(a) \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad (b) \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & p & q \end{bmatrix}$$